

SHLOK CHANNAWAR

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EDUCATION

The Pennsylvania State University
Bachelor of Science in Applied Data Science

State College, PA
December 2027

TECHNICAL SKILLS

Languages: Python (PyTorch, TransformerLens, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib), R, SQL
ML/DL: Neuronpedia Hugging Face (Transformers, Datasets), Weights & Biases
Tools: Git, Jupyter, VS Code, Lambda Cloud, Google Colab

RESEARCH

Look Before You Steer: Geometry Predicts SAE Feature Steerability **ICML MI Workshop**

- **Accepted** to the **ICML Mechanistic Interpretability Workshop** (Seoul); co-first author [\[paper\]](#)
- Proposed and validated a framework showing SAE decoder-space geometry (neighbor density, max cosine similarity) predicts feature steerability before any intervention, achieving Spearman ρ up to **-0.546** ($p < 10^{-6}$) and AUROC up to **0.822**
- Replicated geometry-steerability signal across 4 model architectures (Gemma-2 2B/9B, Llama-3.1-8B, Qwen3-8B) and 2 SAE widths (16K/65K); identified depth-boundary conditions where signal degrades
- Designed end-to-end evaluation pipeline covering 100+ features $\times$ 100 prompts $\times$ 9 steering coefficients with coherence filtering, SALADbench refusal scoring, and permutation null tests
- Optimized batched inference to reduce model calls by 94% (69K $\rightarrow$ 4.3K), cutting GPU compute cost on Lambda Cloud

RLVR for Mathematical Reasoning June 2026 – Present

- Investigating whether training domain-specialist models separately and distilling them produces stronger general math reasoning than training one model on mixed domains directly, using GRPO-style RL with verifiable correctness rewards

Natural Language Autoencoder (NLA) Probing for Contextual Integrity June 2026 – Present

- Investigating whether Qwen2.5-7B internally represents privacy-norm violations before or independent of verbalizing them in output, using linear probes and Natural Language Autoencoders ([NLA](#)) on layer-20 activations across all 4 [ConfAIde](#) tiers
- Found a strong, pre-committed "deflection" signal (AUC 0.89–0.92, decidable before generation) but only a weak "leak" signal (AUC 0.80) — suggesting the model discretely decides whether to deflect but not whether it will leak

Mechanistic Interpretability for AI Financial Advisors (Mechanistic KYC) May 2026 – Present

- Built and validated a rubric-based synthetic client profiler grounded in the Grable & Lytton risk-tolerance scale, generating 6,000 stratified client profiles with full unit-test coverage
- Designing linear probes and activation patching on frozen Gemma-2-9B/Llama-3.1-8B to distinguish belief capture from hypocritical compliance under client pressure, including a novel self-vs-client risk direction decomposition

EXPERIENCE

Technical AI Safety Course June 2026 - Current
BlueDot Impact *Remote*

- Selected for BlueDot Impact's Technical AI safety cohort; studying current alignment techniques, interpretability, and model evaluations through weekly expert-facilitated discussions

Research Assistant January 2025 – June 2025
Humanitarian Engineering and Social Entrepreneurship – Hexarech *State College, PA & Kisumu, Kenya*

- Analyzed sensor data to adjust feed schedules and mitigate poor water conditions, cutting fish mortality 30–50%
- Traveled to Kenya to co-design low-cost demonstration cages with local stakeholders (BMUs, hatcheries, extension officers)